

## Phase 0 — Freeze the Current App (1 day)

Before adding AI, make sure the existing foundation is stable.

- ☒ Expense CRUD
- ☒ Budget CRUD
- ☒ Charts
- ☒ Authentication
- ☒ Expense Categorization ML
- ☒ Dashboard

No new features until this is clean.

---

## Phase 1 — Analytics Engine 🌟 (Highest Priority)

This is the heart of SIA.

**Don't write a single line of LLM code yet.**

Create a backend module:

backend/

analytics/

spendingAnalyzer.js

budgetAnalyzer.js

categoryAnalyzer.js

trendAnalyzer.js

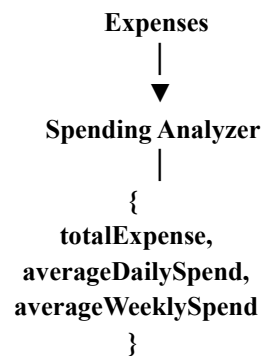
habitAnalyzer.js

healthAnalyzer.js

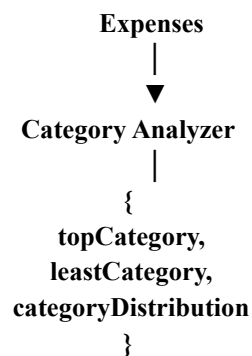
reportGenerator.js

Each analyzer has one responsibility.

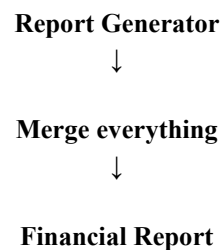
Example:



Another:



Finally:



This report becomes the single source of truth.

---

## Phase 2 — Define the Financial Report

This is probably the most important file in the project.

Something like:

```
{  
  "summary": {},  
  
  "budgets": {},  
  
  "categories": {},
```

```
"trends": {},  
  
"habits": {},  
  
"financialHealth": {},  
  
"forecast": {},  
  
"risk": {}  
}
```

Everything in BALENISA should use this report.

- Dashboard
- Insights
- Notifications
- SIA

Everyone reads from the same report.

---

### **Phase 3 — Forecasting ML**

Now build another ML service.

Input:

Past expenses

Output:

Expected month-end spending

This becomes another section in the report.

---

### **Phase 4 — SIA Backend**

Now create

backend/

sia/

Inside

**intentClassifier.js**  
**contextBuilder.js**  
**promptBuilder.js**  
**llmService.js**  
**responseFormatter.js**

Notice

No frontend yet.

Just backend.

---

## **Phase 5 — Intent Detection**

User asks

How much did I spend on food?

Intent

CATEGORY\_QUERY

User asks

Predict this month's spending

Intent

FORECAST\_QUERY

User asks

Why are my expenses increasing?

Intent

ANALYSIS\_QUERY

Don't involve the LLM if the backend can answer directly.

---

## **Phase 6 — Context Builder**

This is where your backend becomes intelligent.

Example

Question

Why am I overspending?

Context Builder collects

- Budget usage
- Trend analysis
- Category growth
- Forecast

Only these.

Not the whole database.

---

## **Phase 7 — LLM**

Now connect

- OpenAI
- Gemini
- Claude
- Local model

Prompt

**System Prompt**  
+  
**Financial Report**  
+  
**Relevant Context**  
+  
**User Question**

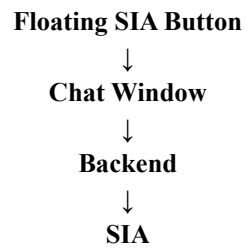
Output

Natural answer.

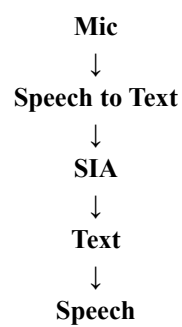
---

## Phase 8 — Frontend Chat

Only now.



## Phase 9 — Voice



## Phase 10 — Smart Notifications

Examples

**Budget Risk**  
**Forecast**  
**Weekend Overspending**  
**Recurring Merchant Alerts**

---

## Final Architecture

